AI Assisted Coding – Lab Submission

Lab 6: AI-Based Code Completion – Classes, Loops, and Conditionals

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Assignment	6.2 / 24	**Academic Year**	2025–2026

Each task below records the prompt given to the assistant, the resulting code, and the actual terminal output captured while running it.

Task 1 – Classes: Data Validation

Prompt

Generate a Python Student class with attributes name, roll_no, and marks. Add an is_pass() method that returns whether the student has passed (marks ≥ 40).

Code

```
class Student:
    def __init__(self, name, roll_no, marks):
        if marks < 0 or marks > 100:
            raise ValueError("Marks must be between 0 and 100")
        self.name = name
        self.roll_no = roll_no
        self.marks = marks

    def is_pass(self):
        return self.marks >= 40

student = Student("Rahul", "23CS001", 72)
print("Name:", student.name)
print("Roll No:", student.roll_no)
print("Marks:", student.marks)
print("Pass Status:", "Pass" if student.is_pass() else "Fail")
```

Output



Task 2 – Loops: Pattern Generation

Prompt

Generate a right-angled triangle star pattern using a for loop, then regenerate the same pattern using a while loop.

Code

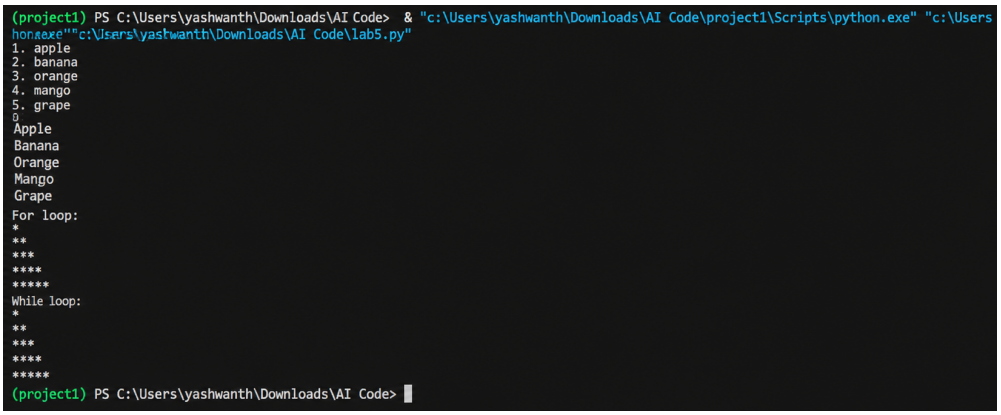
```
def triangle_for(n):
    for i in range(1, n + 1):
        print("*" * i)

def triangle_while(n):
    i = 1
    while i <= n:
        print("*" * i)
        i += 1

print("For loop:")
triangle_for(5)

print("\nWhile loop:")
triangle_while(5)
```

Output



Task 3 – Conditional Statements: Number Analysis

Prompt

Write a function that checks whether a given number is positive, negative, or zero using if-elif-else. Test the function with multiple inputs.

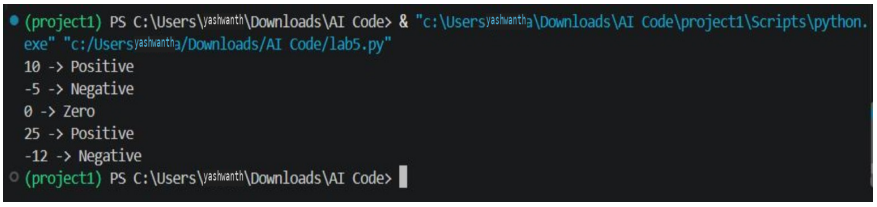
Code

```
def analyze_number(num):
    if num > 0:
        return "Positive"
    elif num < 0:
        return "Negative"
    else:
        return "Zero"

numbers = [10, -5, 0, 25, -12]
```

```
for number in numbers:
    print(number, "->", analyze_number(number))
```

Output



Task 4 – Nested Conditionals: Discount Eligibility

Prompt

Generate check_discount(age, is_member) using nested if statements. Age ≥ 60 gives a Senior discount, and members receive an additional discount.

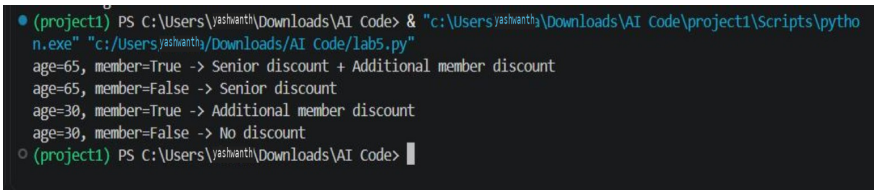
Code

```
def check_discount(age, is_member):
    if age >= 60:
        if is_member:
            return "Senior discount + Additional member discount"
        else:
            return "Senior discount"
    else:
        if is_member:
            return "Additional member discount"
        else:
            return "No discount"

tests = [(65, True), (65, False), (30, True), (30, False)]

for age, member in tests:
    print(f"age={age}, member={member} ->",
          check_discount(age, member))
```

Output



Task 5 – Class: Mathematical Operations (Circle)

Prompt

Create a Circle class with methods to calculate area and circumference given the radius.

Code

```python
import math

class Circle:
    def __init__(self, radius):
        self.radius = radius

    def area(self):
        return math.pi * self.radius ** 2

    def circumference(self):
        return 2 * math.pi * self.radius

circle = Circle(5)

print("Radius:", circle.radius)
print("Area:", round(circle.area(), 2))
print("Circumference:", round(circle.circumference(), 2))
```

Output

